Industrial Internship Report on

Smart City Traffic Forecasting

Prepared by

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Executive Summary
This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT). This internship was focused on a project/problem statement provided by UCT. We had to finish the project including the report in 6 weeks' time. My project was *Smart City Traffic Forecasting System* using Machine Learning to predict vehicle traffic across four city junctions based on historical traffic data. The project involved data preprocessing, exploratory data analysis (EDA), feature engineering, and training a Random Forest Regression model. An interactive Streamlit dashboard was built to allow users to select a junction, date, and time to forecast traffic volume, classify traffic levels (Low, Moderate, High), and provide intelligent traffic insights. The system achieved an R^2 score of approximately 0.969 with a Mean Absolute Error (MAE) of around 2.37, demonstrating high prediction accuracy. This internship gave me a very good opportunity to get exposure to Industrial problems and design/implement solution for that. It was an overall great experience to have this internship.

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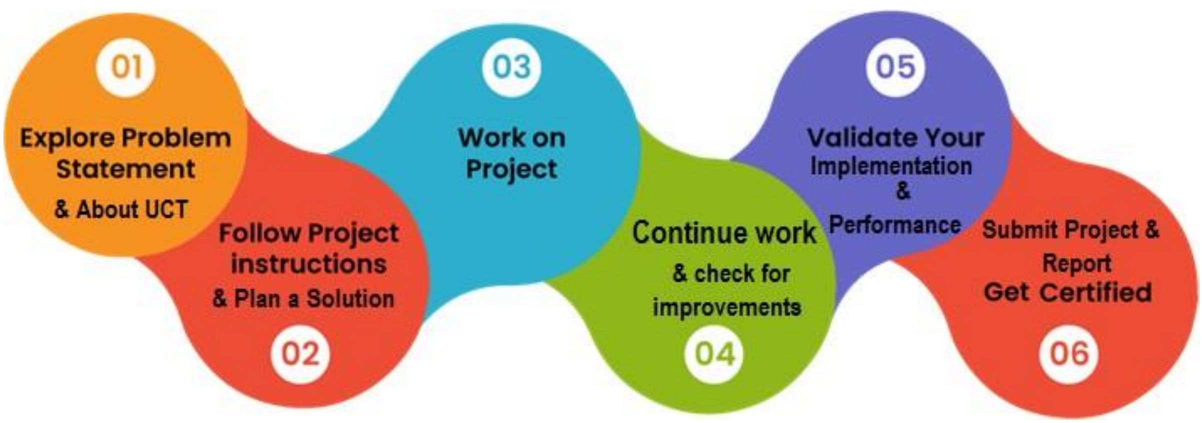
1 Preface

This six-week internship organized by **upSkill Campus**, **The IoT Academy**, and **UniConverge Technologies Pvt. Ltd. (UCT)** provided valuable industry-oriented exposure to Data Science and Machine Learning. Throughout the program, I learned concepts such as Data Science, Probability & Statistics, and Machine Learning while applying them to real-world projects. The primary project, **Smart City Traffic Forecasting System**, involved analyzing historical traffic data and developing a Machine Learning model with a Streamlit dashboard to predict traffic across multiple city junctions. The internship helped bridge the gap between academic learning and practical implementation, strengthening my technical, analytical, and problem-solving skills. The structured weekly learning modules and hands-on project development provided an excellent opportunity to gain practical experience and prepare for future career opportunities in Artificial Intelligence and Data Science.



2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies e.g. Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.

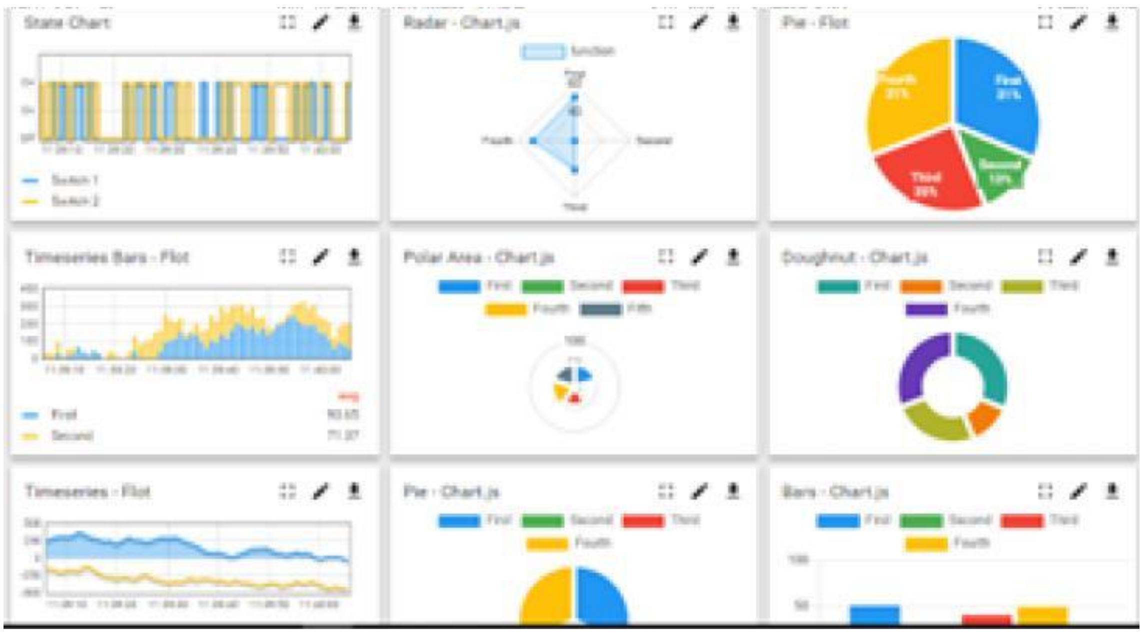
i. UCT IoT Platform (uct Insight)

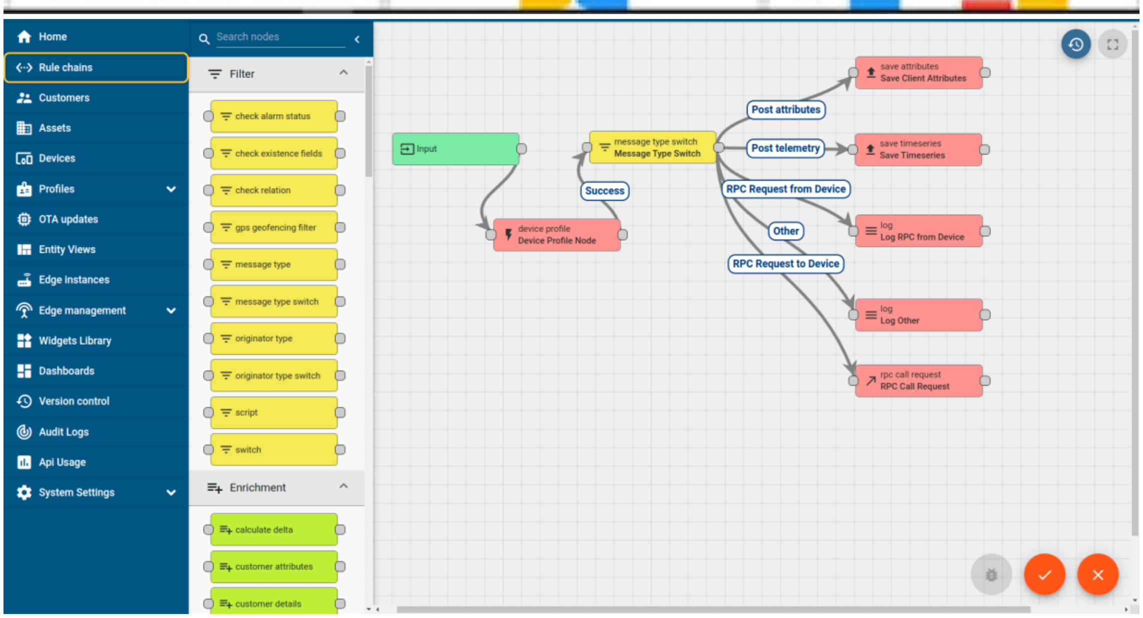
UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable "insight" for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine





ii. Smart Factory Platform (FACTORY WATCH)

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleased the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they what to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.

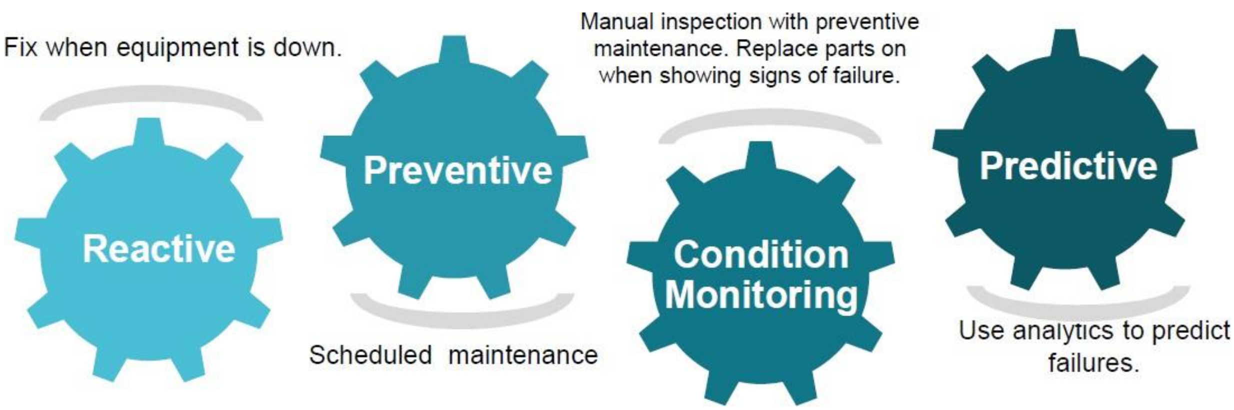
Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i

iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN teschnology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

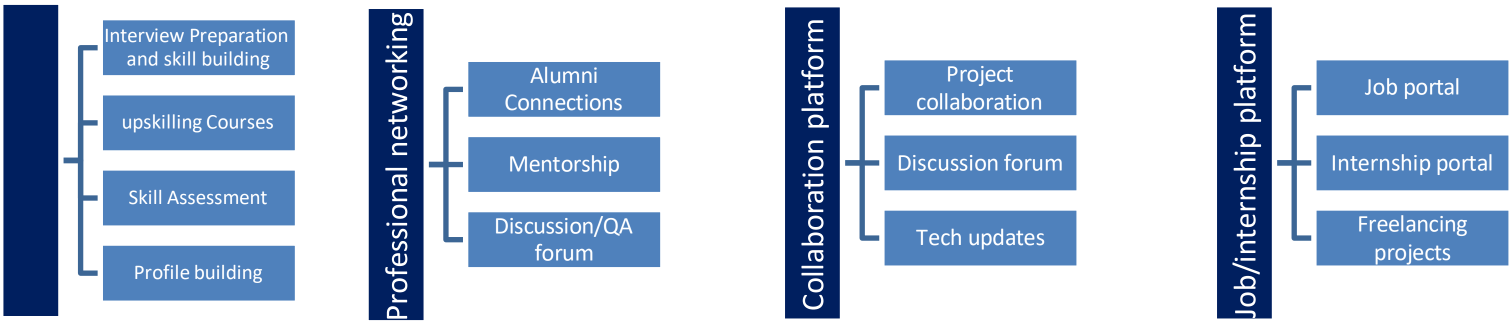
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.



2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to

☛ get practical experience of working in the industry.

☛ to solve real world problems.

☛ to have improved job prospects.

☛ to have Improved understanding of our field and its applications.

☛ to have Personal growth like better communication and problem solving.

2.5 Reference

[1] Breiman, L. (2001). *Random Forests*. Machine Learning, 45(1), 5–32. Available at: https://link.springer.com/article/10.1023/A:1010933404324

[2] Scikit-learn Documentation – RandomForestRegressor – Official documentation for implementing the Random Forest Regression algorithm.

[3] Pandas Documentation – Official documentation used for data preprocessing and manipulation.

[4] NumPy Documentation – Official documentation for numerical computing in Python.

[5] Matplotlib Documentation – Used for data visualization during exploratory data analysis.

3 Problem Statement

The objective of the project was to develop an intelligent Smart City Traffic Forecasting System capable of predicting traffic volume across four major city junctions using historical traffic data. The system considers multiple temporal and contextual factors such as date, time, day of the week, weekends, and public holidays to generate accurate traffic forecasts. By identifying peak traffic periods and traffic congestion patterns in advance, the solution enables city administrators and traffic management authorities to optimize signal timings, improve traffic flow, allocate resources efficiently, and plan preventive measures. The project aims to support data-driven decision-making for smarter urban mobility, reduce traffic congestion, and contribute to the development of sustainable and intelligent smart city infrastructure.

4 Existing and Proposed solution

- **Existing Solution**

Traditional traffic management systems primarily rely on real-time monitoring using CCTV cameras, traffic sensors, and manual supervision by traffic authorities. While these systems provide information about the current traffic conditions, they are mostly reactive in nature and cannot accurately predict future traffic congestion. Existing approaches often require expensive infrastructure, continuous monitoring, and significant human intervention. Additionally, many conventional systems do not effectively utilize historical traffic data to identify recurring traffic patterns or forecast peak traffic periods.

- **Limitations of Existing Solutions**
- Reactive rather than predictive traffic management.
- Limited utilization of historical traffic data.
- High dependence on physical infrastructure such as cameras and sensors.
- Difficulty in forecasting traffic during weekends, holidays, and peak hours.
- Limited decision support for city administrators.

- **Proposed Solution**

The proposed **Smart City Traffic Forecasting System** leverages Machine Learning techniques to predict future traffic volume using historical traffic data collected from four city junctions. The system performs data preprocessing, exploratory data analysis (EDA), and feature engineering to extract meaningful attributes such as hour, day, month, weekday, weekend, holiday, and rush-hour indicators.

A **Random Forest Regression** model is trained to forecast traffic volume with high accuracy. The trained model is integrated into a **Streamlit-based interactive dashboard**, allowing users to select a junction, date, and time to obtain real-time traffic predictions. Based on the predicted traffic volume, the system classifies traffic conditions into different congestion levels and provides intelligent recommendations for traffic management.

- **Value Addition of the Proposed System**
- Predicts future traffic conditions instead of only monitoring current traffic.
- Assists city administrators in planning for traffic peaks.

- Reduces congestion through proactive traffic management.
- Utilizes historical traffic patterns for accurate forecasting.
- Interactive and user-friendly dashboard for quick decision-making.
- Cost-effective solution that can be further integrated with IoT devices and real-time traffic data in future smart city applications.

4.1 Code submission (Github link)

https://github.com/Shad18/upskillCampus

4.2 Report submission (Github link) :

https://github.com/Shad18/upskillCampus/tree/main/Final%20Report

5 Proposed Design/ Model

The proposed **Smart City Traffic Forecasting System** is a Machine Learning-based solution designed to predict future traffic volume across four city junctions using historical traffic data. The system follows a structured workflow beginning with data collection and preprocessing, followed by exploratory data analysis, feature engineering, model training, prediction, and deployment through an interactive dashboard.

The historical traffic dataset is first cleaned and transformed by extracting meaningful temporal features such as **hour, day, month, weekday, weekend indicator, holiday indicator, rush-hour indicator, and season**. These engineered features help the model capture traffic trends and recurring congestion patterns more effectively.

A **Random Forest Regression** algorithm is then trained using the processed dataset. The model learns the relationship between temporal features and traffic volume, enabling it to accurately forecast the expected number of vehicles for a selected junction, date, and time. Model performance is evaluated using **Mean Absolute Error (MAE)** and **R^2 Score**, ensuring reliable prediction accuracy.

The trained model is integrated into a **Streamlit-based web application**, where users can select a junction, date, and time to obtain real-time traffic predictions. The application also classifies the predicted traffic into different congestion levels (Low, Moderate, High, or Peak) and provides visual insights through graphs and interactive dashboards.

5.1 High Level Diagram (if applicable)

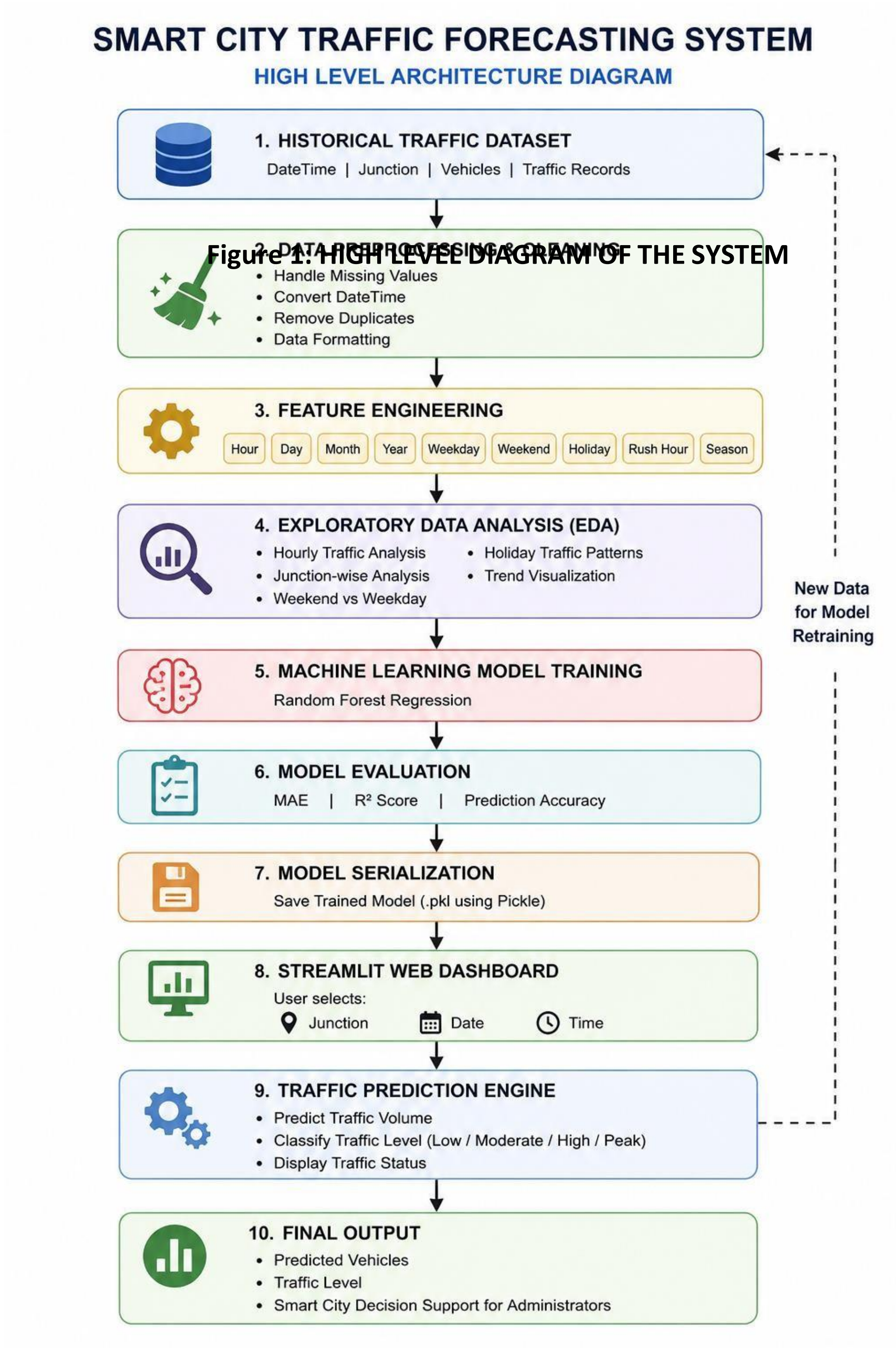


Figure 1: HIGH LEVEL DIAGRAM OF THE SYSTEM

6 Performance Test

6.1 Identified Constraints

The following constraints were considered while developing the Smart City Traffic Forecasting System:

- Prediction accuracy
- Response time
- Memory usage
- Scalability
- Dashboard performance

6.2 Design Considerations

To improve prediction accuracy, feature engineering was performed by extracting Hour, Day, Month, Weekend, Holiday, Rush Hour, and Season features from the DateTime column. A Random Forest Regression model was selected because it handles non-linear traffic patterns effectively.

The trained model was saved as a Pickle file, allowing the Streamlit dashboard to load it once during startup and generate predictions almost instantly. Only essential features were used, reducing memory consumption and improving execution speed.

6.3 Performance Results

Metric	Result
Algorithm	Random Forest Regression
Mean Absolute Error (MAE)	**2.38**
R^2 Score	**0.969**
Prediction Time	**< 1 second**
Dashboard	Streamlit Web Application

The model achieved an **R^2 score of 0.969**, indicating high prediction accuracy, while the **MAE of 2.38** shows that prediction errors are minimal. The application provides fast and reliable predictions suitable for real-time traffic analysis.

6.4 Limitations

The current system relies only on historical traffic data and does not consider live weather conditions, accidents, road construction, or public events. Additionally, it currently supports predictions for only four traffic junctions.

6.5 Future Enhancements

The system can be further improved by integrating real-time traffic sensors, weather APIs, accident detection, IoT devices, and advanced deep learning models such as LSTM. Cloud deployment and support for additional city junctions would make the solution more suitable for large-scale smart city applications.

7 Performance Outcome

The Smart City Traffic Forecasting System successfully met the primary objective of accurately predicting vehicle traffic across four city junctions using historical traffic data. By applying feature engineering techniques and a Random Forest Regression model, the system learned traffic patterns based on date, time, weekdays, weekends, holidays, and seasonal variations.

The final model achieved a **Mean Absolute Error (MAE) of approximately 2.38 vehicles** and an **R^2 Score of 0.969**, demonstrating excellent prediction accuracy and strong generalization on unseen data. These results indicate that the model can explain nearly **97% of the variation** in traffic volume, making it suitable for practical traffic forecasting applications.

The Streamlit-based dashboard provides an intuitive interface where users can select the junction, date, and time to obtain instant traffic predictions. Along with the predicted vehicle count, the application categorizes traffic into different congestion levels such as **Low, Moderate, High, and Very High**, allowing users to quickly interpret traffic conditions.

The system also includes intelligent feature extraction, automatically identifying weekends, holidays, rush hours, and seasons without requiring manual input. This significantly improves prediction quality compared to using only raw date and time values.

Overall, the project demonstrates that Machine Learning can effectively support smart city traffic management by forecasting congestion, assisting city administrators in resource planning, reducing delays, and improving traffic flow. The developed solution is scalable and can be enhanced further by integrating live traffic feeds, weather information, IoT sensor data, and deep learning models such as LSTM for even higher prediction accuracy.

Key Performance Metrics

- **Machine Learning Algorithm:** Random Forest Regression
- **Mean Absolute Error (MAE): 2.38**
- **R^2 Score: 0.969**
- **Prediction Time:** Less than **1 second**
- **Supported Junctions:** 4
- **User Interface:** Streamlit Dashboard
- **Overall Outcome:** High prediction accuracy with fast and reliable real-time traffic forecasting suitable for smart city applications.

8 My learnings

During this internship, I gained practical knowledge in **Data Science, Probability & Statistics, Machine Learning, and Model Deployment** by working on real-world projects. I learned the complete machine learning workflow, starting from data preprocessing and exploratory data analysis (EDA) to feature engineering, model training, evaluation, and deployment.

While developing the **Smart City Traffic Forecasting System**, I learned how to extract meaningful features from date and time data, build regression models using **Random Forest Regression**, evaluate model performance using **MAE** and **R^2 Score**, and improve prediction accuracy through feature engineering. I also learned how to save trained models using Pickle and deploy them using a **Streamlit** web application.

In addition, I strengthened my programming skills in **Python** and gained hands-on experience with important libraries such as **Pandas, NumPy, Matplotlib, Scikit-learn, and Streamlit**. I also improved my understanding of data visualization, model optimization, and GitHub project management.

The internship enhanced my analytical thinking, debugging abilities, and problem-solving skills while giving me confidence in building complete end-to-end machine learning applications. Overall, this internship provided valuable industry-oriented experience and strengthened my readiness to work on future AI and Data Science projects.

9 Future work scope

The Smart City Traffic Forecasting System provides a strong foundation for intelligent traffic management; however, several enhancements can further improve its functionality and practical applicability.

Future work includes integrating **real-time traffic data** from IoT sensors, GPS devices, and traffic cameras to provide live traffic predictions. Incorporating **weather information**, road construction updates, accidents, and public event data can further improve forecasting accuracy by considering external factors that influence traffic conditions.

The current Random Forest Regression model can also be extended using advanced **Deep Learning techniques** such as Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) networks for better time-series forecasting. Additionally, deploying the application on cloud platforms would improve accessibility and support multiple users simultaneously.

The system can be expanded to monitor traffic across multiple cities, generate congestion heatmaps, optimize traffic signal timings, and provide route recommendations. These enhancements would transform the application into a comprehensive smart traffic management solution capable of supporting real-time decision-making for modern smart city infrastructure